

Paper Notes/Structure

Wednesday, July 1, 2026 1:36 PM

History of SLAM

Go here for more historic context:

Fast and Accurate Dense Scene Reconstruction with Learning-Based Visual SLAM and Radiance Fields
2026 PhD Thesis

COLMAP

SIFT/ORB/Third One

SuperGlue/SuperPoint

SLAM

Monocular

Sensor Based Approaches

Online VS Offline

Modern Approach's

Gaussian Splats

Sensor Fusion

Reduction down to monocular or RGB-D as the field progresses

Simultaneous emergence of NeRF approaches

Positives of NeRF

Cons of NeRF

3D Gaussian Splatting Introduced

Shown to have better results than NeRF approaches.

Incorporation of prior and learned priors

Your could an in a historical component that looks at the feed forward approaches that lead to the field being ready for the shift provided by M3/D3.

Ex: [Dynamic Visual SLAM using a General 3D Prior](#)

VGT and Dino basis. Interesting multihead output controlled by a self attention layer.

Core to the ground breaking approaches of M3STER/DUST3R

DUST3R

Rather than using interest points and descriptors this used a trained transformer to generate a 3D dense pointmap from two cross.

The prior is that each pixel in the source images can be used to meaningfully generate 3D points. Other priors could include: flat walls, smooth surfaces, manhattan style features, etc.

Unlike COLMAP you also don't need to worry about camera calibration. Big plus! But Colmap is still considered faster.

This gap is being closed however. It looks like DUST4R/MASt4R/etc approaches are highly graphic intensive though.

At their core these are tranformer based approaches.

MASt3R improves on this by creating a matching feature map along with the point map.

[MASt3R-SLAM: Real-Time Dense SLAM with 3D Reconstruction Priors](#)

The head that creates this was custom trained for this implementation (HeadDesc). Training to create the feature vectors was done on 14 datasets: Habitat, ARKitScenes, BlendedMVS, MegaDepth, Static Scenes 3D, ScanNet++, CO3D-v2, Waymo, Map-free, WildRGB, VirtualKitti, Unreal4K, TartanAir, and an internal dataset.

The goal of training was to create feature vectors that were useful for matching.

MASt4R argues DUST4R was not explicitly trained for matching.

The structure of the resulting feature vectors was not specified prior to training outside of the n=24 size. The content and pattern was purely determined by matching reward contrastive learning.
(InfoNCE-style contrastive loss)

Tangents

4D & 5D SLAM

GOLMAP

FASTMAP

The challenges of our particular environment

High visual similarity/Manhattan type environments

Monocular depth perception

Loop Closure in low visual density environments

Trade of information density for speed

Speed: [Faiss](#) A library for efficient similarity search and clustering of dense vectors.

Rankings of best approaches: <https://chatgpt.com/g/g-p-6a3989f5909c8191b48522794c68aea3-umbc/c/6a45bd68-1dd8-83ea-86e5-db3ffa9cf704>

Proposed solutions

Incorporation of semantic landmarks

High pressure on bundle adjustments to account for high drift likelihood

Incorporation of M3STER/DUST3R approach

Global bundle requirements

Feed Forward Approach?

Paper writing approach

Propose our approach as recognizing a long standing issue in application of visual SLAM to repetitive office/university spaces.

List prior examples highlighting the issue.

Working in prior approaches somewhere to.

Our aim is to extend this by testing modern models in a realistic university environment.

Maybe have 3 separate categories? Highly decorated. Medium decorated. Bare.

An extended application is hospitals & healthcare.

Semantic Approaches

Keystone Paper

Introduced the term semantic slam ([Link](#))

First implementation ([Link](#))

Semantic Segmentation (Voxel/Pixel)

Why not apply a bag-of-words approach to objects instead of interest points?

PanSt4R

Monocular RGB as input. Latest proven approach in Semantic category.
X, X, and X improvements over COLMAP, M4STER, DUST3R

If this works well we can then explore the efficacy of PanSt4R's (2025) successors.

[Pano3D](#)

Improvement on PanSt4R and an intermediate model [SIU3R](#)

[UNITE](#)

Tangential development but states an improvement over PanSt4R.

Incorporates an open vocabulary search (Large vocab)

What is the limit to this? Does it inform localization?

Tangentially developed approaches

[IGGT](#)

[Uni3R](#)

[VGGT-Segmentor](#)

[EPS3D](#)

Text Aware SLAM (Key Competing Approach)

Good for offices

[TextSLAM: Visual SLAM with Semantic Planar Text Features](#)

Repetitive Spaces

Negative Examples (these struggled)

[Probabilistic Data Association for Semantic SLAM](#)

Tested on a 175 m trajectory around one floor of an office building and a 625 m trajectory around two floors of an office building. In the second experiment, the semantic classes include red office chairs and doors.

The paper explicitly identifies the office hallway problem: the hallways were extremely repetitive, bag-of-words loop closure produced false positives, and ORB-SLAM2 failed because of false loop closures.

A lot of this seems oriented towards loop closure

[Language-Extended Indoor SLAM \(LEXIS\): Versatile System for Real-time Visual Scene Understanding](#)

Utilizes the open vocabulary of LLMs to create the high level hierarchical labels for rooms.

How is this done? An LLM or MLLM can't read a 3D space.

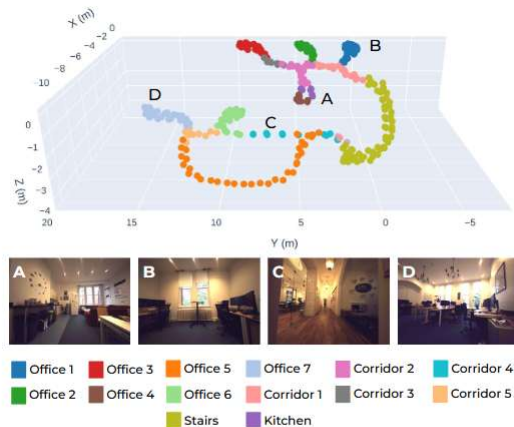


Fig. 1: LEXIS enables pose graph segmentation from natural language. By exploiting the open-vocabulary capabilities of CLIP, we can segment room instances such as office, kitchen, and corridor directly from the pose graph without fine-tuning. The above dataset is from a two floor office environment and contains 7 rooms as well as 2 corridors and stairs.

Interesting Alternative Approaches

LLM Incorporation

[Semantic Enhancement for Object SLAM with Heterogeneous Multimodal Large Language Model Agents](#)

Room-level / topological semantic SLAM

Potentially useful for expanding the size and scope of the project past a single floor

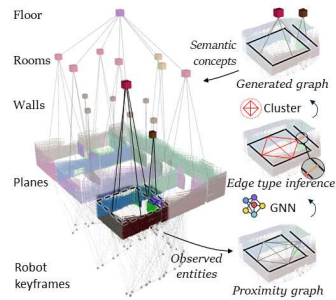
Simple Start: [Marker-based Visual SLAM leveraging Hierarchical Representations \(Paper\)](#) (2024)

Visual markers (QR Codes) to differentiate walls, rooms, corridors.

Show to be an improvement over just regular visual marker SLAM.

One of the first: [Topomap: Topological Mapping and Navigation Based on Visual SLAM Maps](#) (2018)

Most have the same base structure



[Learning High-level Semantic-Relational Concepts for SLAM](#)

[S-Graphs 2.0 – A Hierarchical-Semantic Optimization and Loop Closure for SLAM](#)

[SLAM-Free Visual Navigation with Hierarchical Vision-Language Perception and Coarse-to-Fine Semantic Topological Planning](#)

This one goes further (read more) It labels furniture.

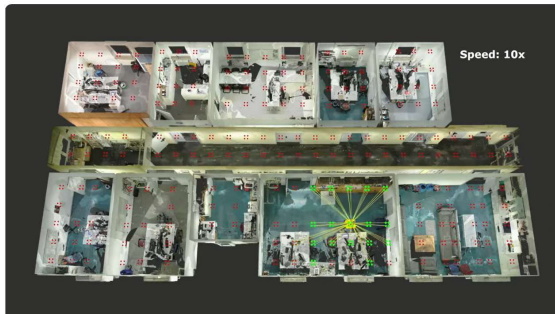
[Semantic Mapping in Indoor Embodied AI – A Comprehensive Survey and Future Directions](#)

Interesting But Not Proposed

QR Code Based Approaches (Add a few more examples here)

[Benchmark for Evaluating Long-Term Localization in Indoor Environments \(IROS'25\)](#)

Uses many many codes for determining precise location not just room or object



Collaborative SLAM

[ICRA 2023 - Cross-Agent Relocalization for Decentralized Collaborative SLAM](#)

Dynamic SLAM (4D SLAM)

It can also be semantic! (5D SLAM...)

[SDD-SLAM: Semantic-Driven Dynamic SLAM With Gaussian Splatting](#)

Loop Closure

[ROVER: Robust Loop Closure Verification with Trajectory Prior in Repetitive Environments](#) (2025)

Focused on visual SLAM. Algorithm tested on master and duster.

[Robust Loop Closure by Textual Cues in Challenging Environments](#) (2024)

Title says exactly what it does.

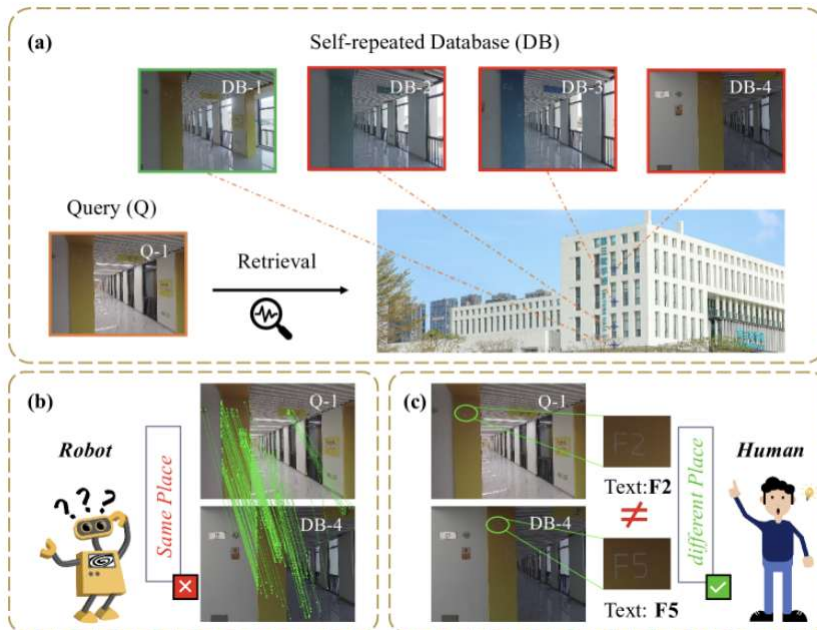
VPR

Sub component within SLAM

This can be a key discriminator in confusing places.

Add as a branch?

[TextInPlace: Indoor Visual Place Recognition in Repetitive Structures with Scene Text Spotting and Verification](#) (2025)



Privacy

Technologies/Pipeline

Further Research Approaches to Explore

Method routing

Inspired by current challenges in LLM costs, can system functionality be optimized by detecting and employing different techniques based on perceived confidence of success?

Ex: COLMAP is incredibly fast but bad with hallways.

Other approaches such as ___, ___, ___ have improved performance but require additional computational resources. Beyond what is available on mobile devices.